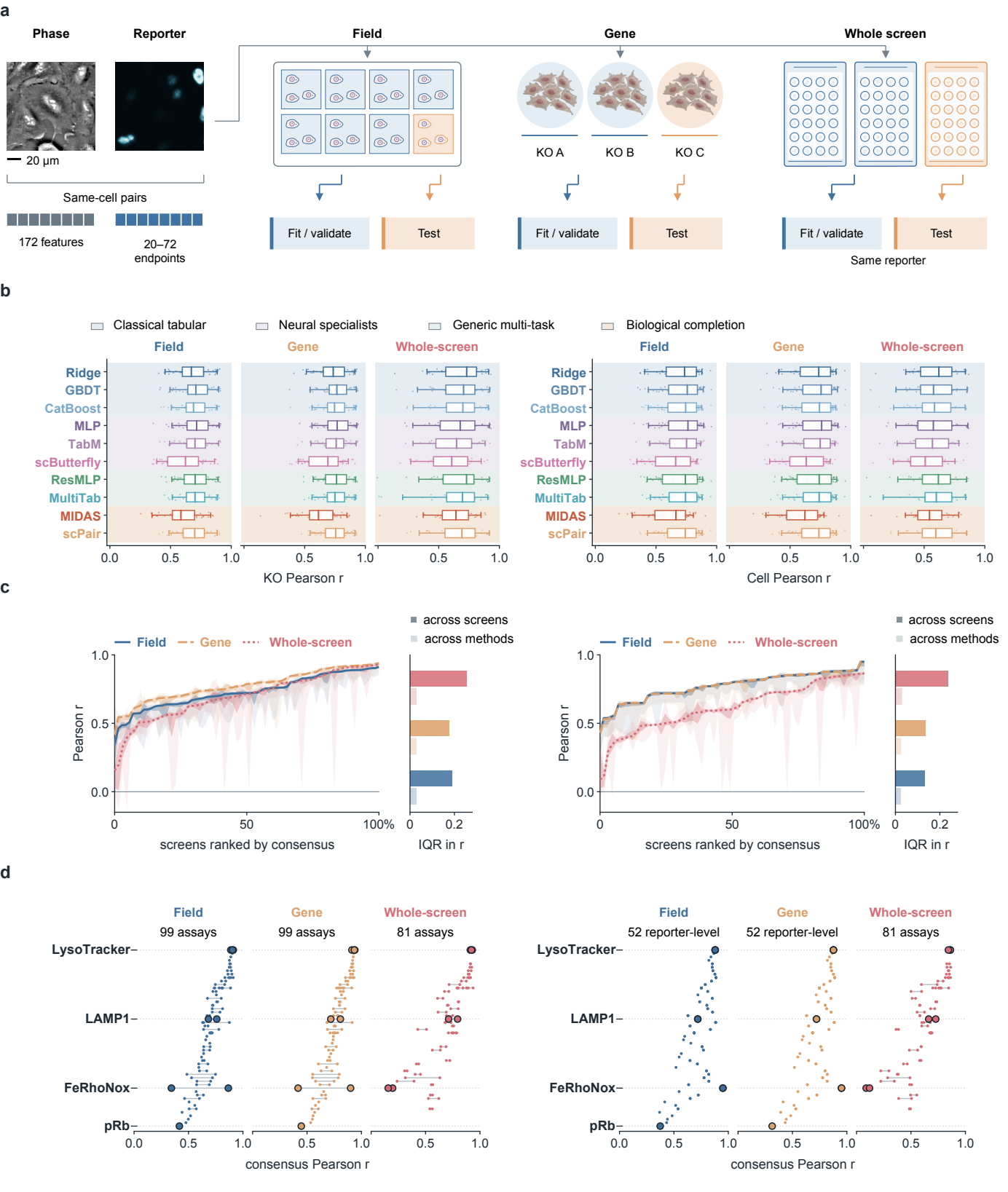

a
Phase
Reporter
20 µm
Same-cell pairs
172 features
20–72 endpoints
Field
Gene
Whole screen
KO A
KO B
KO C
Fit / validate
Test
Fit / validate
Test
Fit / validate
Test
Same reporter
b
Classical tabular
Neural specialists
Generic multi-task
Biological completion
Field
Gene
Whole-screen
Ridge
GBDT
CatBoost
MLP
TabM
scButterfly
ResMLP
MultiTab
MIDAS
scPair
0.0
0.5
1.0
KO Pearson r
Cell Pearson r
c
Field
Gene
Whole-screen
across screens
across methods
Pearson r
screens ranked by consensus
IQR in r
d
Field
Gene
Whole-screen
99 assays
99 assays
81 assays
52 reporter-level
52 reporter-level
81 assays
LysoTracker
LAMP1
FeRhoNox
pRb
consensus Pearson r